Alberta's variance paperwork keeps bouncing back

- 1,553 / 637 / 1,698 / 1,713 building variances in 2019–2022
- Edmonton: about 67% of 2023 variances (all disciplines)
- No published service standard for variance review

Sources: SCC Annual Report 2023; challenge scope

Describe your project in plain words [Intake, 20]

- Your description becomes facts, each tagged with where it came from **[if merged: PR #16]**
- Designed so applicant and site details come only from you; upload isn't available yet
- "Not sure" becomes an open question instead of a dead end (in review, PR #16)

EXAMPLE DATA



Screenshot: PR #16 Describe screen with pasted case C (homebuilder) text, authority set to Edmonton, Review gaps listing owner, mailing address and legal description as missing

PR #16 screen (in review)

`shot-s02-describe`

Sources: dev `engine/models.py`, `engine/completeness.py`; PR #16

It asks before it guesses, and we test that [Intake, 20]

- Missing fields become plain questions, each tied to a form section (engine on dev; screen in review)
- Engineer-needed flag (PR #8, on dev): rules_only 55/60 in-sample, 1 under-flag; shipped rules_plus_unknown 50/60 in-sample, 0 under-flags (on text-only cases it answers only yes or ask, never no, so 0 under-flags holds by construction), 27 unknowns sent to the applicant; held-out 7/8 blind (pre-fix), post-fix 4/8 (no longer blind); model-checked, not human-verified. (rules_only is the rules-only checker.)
- Accuracy across all applicant types: still to be measured

EXAMPLE DATA



Screenshot: PR #16 Code check screen with the manufacturer's two “Your words” cards, both answered Not sure and logged as open questions

PR #16 screen (in review)

`shot-s03-not-sure`

Sources: dev `engine/completeness.py`; PR #8

A ranked shortlist, measured against a named acceptable solution [Code Identification, 20]

- Ranked articles with a named benchmark acceptable solution **[if merged: PR #12]**
- Identify never adds an article on its own; you confirm each one. p ≥ 0.75 only sets a "suggested" hint (uncalibrated) **[if merged: PR #12]**; the command-line tool on dev still auto-commits Part 9 articles
- Section C objectives and functions are looked up from the code's tables where a row exists; otherwise the screen says the lookup is not available (on dev); the table stays empty until an article is added from Search, because identification is command-line only

EXAMPLE DATA



Screenshot: PR #16 Code search for the bedroom window, the returned row added to the check, and its Section C row looked up from the attribution tables

PR #16 screen (in review)

`shot-s04-search`

EXAMPLE DATA



Still: PR #12 identify output and benchmark card

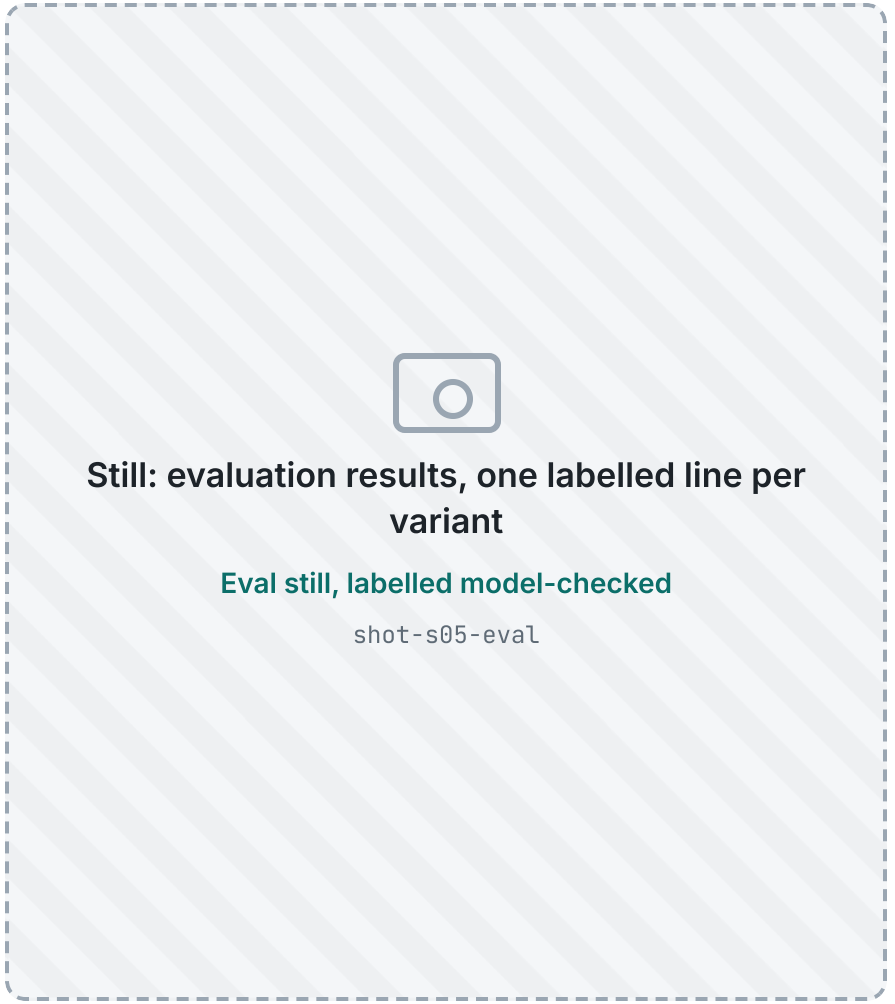
PR #12 still, labelled ‘not in this app build’

`shot-s04-benchmark`

Sources: dev `engine/section_c.py`, `engine/identify/`; PR #12; case C (PR #14, on dev)

Every citation resolves, or it's dropped [Code Identification, 20]

- Built by script from the official NBC(AE) 2023; the codedb README on dev names the 2nd printing; the SHA-256 pin is in draft PR #12
- A citation that doesn't quote the code word for word is removed (on dev)
- Identification (model aliases, not pinned): C$0.0277 per case, 5.7 s median, 11.3 s p95 (measured, 20 dev cases, cost and latency over 3 repeats, identification step only (excludes the completeness judge and Section D))



Still: evaluation results, one labelled line per variant

Eval still, labelled model-checked

`shot-s05-eval`

Sources: Alberta building codes; draft PR #12 (keyed run, Oct 20, 2026; accuracy from one run only); Implementation Plan §9

It knows where to stop [Recommendations, 10]

- Lists the evidence an alternative solution needs, in order (engine on dev; screen in review)
- Says when a registered professional is required, and cites the Division C rule (e.g. 2.4.1.1.(3), 2.4.1.2.(1)) when it applies (engine on dev)
- Never writes engineering judgment itself

EXAMPLE DATA



Screenshot: PR #16 Alternative path with its six fields left empty and the Section D card stating the tool does not write the evidence or the professional opinion

PR #16 screen (in review)

`shot-s06-section-d`

Sources: dev `engine/section_d.py`, `engine/professional.py`; IFC engineering-judgment guidelines; case B (PR #14, on dev)

Completeness review: built in the engine, officer screen in review [Review, 10]

- Gaps listed by form section A–D; completeness only, never compliance (engine on dev)
- Officer screen with a drafted information request **[if merged: PR #16]**
- Div C 2.3.1 and one-location gates, portal package **[if merged: PR #11]**

EXAMPLE DATA

Screenshot: PR #16 Officer review screen with one planted sample package, before and after fixes, and Copy request

PR #16 screen (in review)

`shot-s07-officer`

Sources: IB-2025-11-18; IB-2026-10-01; dev `engine/completeness.py`

One tool, three kinds of first-time applicants [General Applicability, 5]

- Homebuilder: slab edges need detailing, the unvented roof probably fails its STANDATA as drawn, the window needs a variance and fall protection **[case files: PR #14, on dev]**
- Manufacturers can seek a voluntary CCMC assessment (or another recognized evaluation) as evidence for the authority having jurisdiction, which makes the approval decision; a site-specific variance covers only that project **[case files: PR #14, on dev]**
- General contractor: plastic pipe needs 50 Pa-tested firestops, Clause 3.1.9.4.(4)(b) **[case files: PR #14, on dev]**

EXAMPLE DATA



Screenshot: PR #16 “Who are you?” role picker: Manufacturer, General contractor, Homebuilder

PR #16 screen (in review)

`shot-s08-roles`

Sources: STANDATA 23-BCV-017; STANDATA 23-BCV-001; STANDATA 23-BCV-003; Calgary B23-014

Fills the current SCC form; built to travel [Usability and Portability, 10]

- Fills Part 1, Sections A–D of the Council's current SCC form (PDF last modified Oct 6, 2026), not the brief's older Appendix 1 template; same field names, differences listed in the Implementation Plan (the engine fills the SCC form; in the app, PR #16 in review, Check completeness downloads it unsigned, officer Part 2 blank, once every gap is closed)
- The engine blocks export with reasons until gaps clear, or returns a labelled draft (engine on dev; in the app (PR #16, in review), once every gap is closed, Check completeness downloads the Council's October 2026 variance form, filled in by the engine, unsigned, officer Part 2 blank, with a notice that it is not a finding that the design complies; Download draft record separately saves a .txt draft record)
- Designed so local add-ons are configuration; per-jurisdiction form registry in PR #12

EXAMPLE DATA



Screenshot: PR #16 Describe authority select, then Application with Check completeness disabled while gaps are open

PR #16 screen (in review)

`shot-s09-locked`

EXAMPLE DATA



Screenshot: the downloaded, engine-filled SCC variance form PDF (unsigned, officer Part 2 blank); the 'not a finding that the design complies' notice is in the app's dialog, not the PDF; the PDF appendix says it 'does not state that the proposed design complies'

PR #16 (in review): Check completeness download, AI-agent simulated session

`shot-s09-scc-form`

Sources: dev `engine/export/field_maps/scc_2026-10.json`, `engine/export/scc.py`; SCC form; PR #12

Ready for December 1 [Implementation Plan, 10]

- From Dec 1, 2026, the SCC portal is the only way to register a variance, and registration is not approval
- The authority having jurisdiction is the organization, so we pilot with an organization
- Oct 29 Submission 1 (demo build) → Nov 26 Submission 2, pilot build in a Canadian region (our plan) → Dec 2 – Mar 2 pilot window (not guaranteed)

Sources: IB-2026-10-01; IB-2026-10-16; onboarding guide

Applicants pay to arrive complete [Business Plan, 10]

- First buyer: production homebuilders, who write the Div C 2.3.1 documentation
- Price to test: C$125–750 per bespoke package (untested hypothesis)
- Identification model cost (model aliases, not pinned): C$0.0277 per case, about 0.02% of the C$125 floor (measured on dev cases; identification step only)
- Defensible as models improve: form-version tracking, the Section C attribution layer with its citation validator, and a place in the officer's workflow (Judge Q&A 13)

Sources: draft PR #12 (keyed run, Oct 20, 2026); SCC fee schedule; Edmonton Procurement Standard; Calgary procurement policy

We read the regulator's own data [X Factor, 5]

- Caught the Council's Oct 6, 2026 form change; export pinned to it
- 1,501 of 3,088 building article entries (Q1–Q2 2026) are one soffit article, 9.10.15.5.(8)
- Designed around three SCC bulletins from the past year
- No code-AI product we reviewed advertises officer-side alternative-solution review

Sources: SCC Data Hub; Business Plan §7